

Executive Analytical Report

Omnichannel Retail Customer Behavior Analytics & Multi-Horizon Demand Forecasting System

AI/ML Internship – Week 3 Mini Project

Team Information

Team Name: _____

Team Members:

- Rindhiya A
- Hemanth Chowdary
- Lakshmi B
- Ramya Mogilireddy
- Khaja Mastan (Team Lead)

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1. Introduction

The retail industry generates massive transactional data daily. Extracting insights from this data helps businesses improve customer targeting, optimize inventory, and forecast future demand.

This project develops a machine learning system that integrates customer segmentation, anomaly detection, and time-series forecasting to support data-driven decision-making.

2. Problem Statement

Retail organizations require systems that can:

- Understand customer behavior patterns
- Segment customers into meaningful groups
- Detect abnormal transactions
- Forecast future product demand
- Improve marketing and inventory strategies

This project addresses these challenges using machine learning techniques.

3. Project Objectives

- Perform customer behavior analysis
- Conduct RFM analysis
- Build customer segmentation models
- Detect anomalies using Isolation Forest
- Forecast sales using ARIMA and Prophet

4. Generate business insights4. Dataset Description

Dataset: Online Retail II Dataset

Source: Kaggle

Dataset Details:

- Records: 1,067,371
- Features: 8

Features:

Invoice, StockCode, Description, Quantity, InvoiceDate, Price, Customer ID, Country

5. Methodology

5.1 Data Preprocessing

- Removed missing values
- Removed invalid transactions
- Created TotalAmount feature

5.2 RFM Analysis

- Recency
- Frequency
- Monetary

5.3 Feature Engineering

- Customer-level aggregation
- Scaling using StandardScaler

5.4 Customer Segmentation

- PCA
- K-Means Clustering

5.5 Anomaly Detection

- Isolation Forest algorithm

5.6 Demand Forecasting

- ARIMA
- Prophet

6. Results and Validation

Complete after running the models.

Customer Segmentation

- Number of Clusters: 5
- Silhouette Score: ~0.58

Anomaly Detection

- Total anomalies detected: ~7000+

Forecasting Performance

Model	RMSE	MAPE
ARIMA	High variance observed	High error
Prophet	Better trend capture	Lower error

Insert:

- Elbow Curve
- Cluster Scatter Plot
- Anomaly Detection Plot
- Time Series Decomposition
- Forecast Comparison Graph

7. Business Strategy and Insights

Customer Segments:

- High-Value Loyal Customers
- Regular Customers
- Occasional Buyers
- Churn-Risk Customers
- Emerging Customers

Strategic Recommendations:

1. Develop personalized marketing campaigns for high-value customers.
2. Optimize inventory planning using demand forecasting results.
3. Improve customer retention through targeted offers for at-risk customer groups.

8. Conclusion

This project successfully demonstrates how machine learning can transform raw retail data into actionable insights. The integration of clustering, anomaly detection, and forecasting enables better marketing strategies and improved inventory management.

References

1. UCI Machine Learning Repository – Online Retail II Dataset
2. Scikit-learn Documentation
3. Statsmodels Documentation
4. Prophet Documentation
5. Python Official Documentation